Mat 1575 HW6 (§3.1)

Find the integral by using the simplest method. Not all problems require integration by parts.

7. $\int \ln x\,dx$ $= x\ln x - \int x\frac{dx}{x}$

$u=\ln x \quad dv=dx$

$du=\frac{dx}{x} \quad v=x$

$= x\ln x - \int dx$

$= x\ln x - x + C$

8. $\int x\cos x\,dx$ $= x\sin(x) - \int \sin(x)\,dx$

$u=x \quad dv=\cos(x)dx$

$du=dx \quad v=\sin(x)$

$= x\sin(x) + \cos(x) + C$

13. $\int xe^{-x}\,dx$ $= -xe^{-x} - \int -e^{-x}\,dx$

$u=x \quad dv=e^{-x}dx$

$du=dx \quad v=-e^{-x}$

$= -xe^{-x} + \int e^{-x}dx$

$= -xe^{-x} - e^{-x} + C$

15. $\int x^2\cos x\,dx$ $= x^2\sin(x) - \int 2x\sin(x)\,dx = x^2\sin(x) - \left[-2x\cos(x) - \int 2\cos(x)\,dx\right]$

$u=x^2 \quad dv=\cos(x)dx$

$du=2xdx \quad v=\sin(x)$

$u=2x \quad dv=\sin(x)dx$

$du=2dx \quad v=-\cos(x)$

$= x^2\sin(x) + 2x\cos(x) + \int 2\cos(x)dx$

$= x^2\sin(x) + 2x\cos(x) + 2\sin(x) + C$

16. $\int x\ln x\,dx$ $= \frac{x^2}{2}\cdot\ln x - \int \frac{x^2}{2}\frac{dx}{x} = \frac{x^2}{2}\ln x - \int \frac{x}{2}\,dx$

$u=\ln x \quad dv=x\,dx$

$du=\frac{dx}{x} \quad v=\frac{x^2}{2}$

$= \frac{x^2}{2}\ln x - \frac{x^2}{4} + C$

19. $\int e^x\sin x\,dx$ $= e^x\sin(x) - \int e^x\cos(x)\,dx = e^x\sin(x) - \left[e^x\cos(x) - \int -e^x\sin(x)\,dx\right]$

$u=\sin(x) \quad dv=e^x dx$

$du=\cos(x)dx \quad v=e^x$

$u=\cos(x) \quad dv=e^x dx$

$du=-\sin(x)dx \quad v=e^x$

$= e^x\sin(x) - e^x\cos(x) - \int e^x\sin(x)\,dx$

$\Rightarrow \int e^x\sin(x)\,dx = e^x\sin(x) - e^x\cos(x) - \int e^x\sin(x)\,dx$

$\Rightarrow 2\int e^x\sin(x)\,dx = e^x\sin(x) - e^x\cos(x) + C \Rightarrow \int e^x\sin(x)\,dx = \frac{1}{2}\left[e^x\sin(x) - e^x\cos(x)\right] + C$

20. $\int e^x\cos x\,dx = \frac{1}{2}\left[e^x\sin(x) + e^x\cos(x)\right] + C$ (check classwork 07, item 9)

27. $\int x^2 \ln x \, dx$

$= \frac{x^3}{3}\ln x - \int \frac{x^3}{3}\frac{dx}{x} = \frac{x^3}{3}\ln x - \int \frac{x^2}{3}dx$

$= \frac{x^3}{3}\ln x - \frac{x^3}{9} + C$

$u = \ln x \quad dv = x^2 dx$

$du = \frac{dx}{x} \quad v = \frac{x^3}{3}$

31. $\int x^2 \sin x \, dx$

$= -x^2\cos(x) - \int -2x\cos(x)dx = -x^2\cos(x) + \int 2x\cos(x)dx$

$u = x^2 \quad dv = \sin(x)dx$

$du = 2x dx \quad v = -\cos(x)$

$u = 2x \quad dv = \cos(x)dx$

$du = 2dx \quad v = \sin(x)$

$= -x^2\cos(x) + 2x\sin(x) - \int 2\sin(x)\,dx$

$= -x^2\cos(x) + 2x\sin(x) + 2\cos(x) + C$

Compute the definite integrals. Use a graphing utility to confirm your answers.

38. $\int_{1/e}^{1} \ln x \, dx$

$= \left[x\ln x\right]_{\frac{1}{e}}^{1} - \int_{\frac{1}{e}}^{1} x\frac{dx}{x} = \left[1\ln(1) - \frac{1}{e}\ln\left(\frac{1}{e}\right)\right] - x\Big|_{\frac{1}{e}}^{1}$

$= \left[0 - \frac{1}{e}\cdot(-1)\right] - \left[1 - \frac{1}{e}\right] = \frac{1}{e} + 1 + \frac{1}{e} = \frac{2}{e} - 1$

$u = \ln x \quad dv = dx$

$du = \frac{dx}{x} \quad v = x$

$\ln(1) = 0$

$\ln\left(\frac{1}{e}\right) = \ln(e^{-1}) = -\ln(e) = -1$

42. $\int_0^{\pi} x\cos x \, dx$

$= \left[x\sin(x)\right]_0^{\pi} - \int_0^{\pi}\sin(x)\,dx = \left[x\sin(x)\right]_0^{\pi} + \left[\cos(x)\right]_0^{\pi}$

$= \left[\pi\sin(\pi) - 0\cdot\sin(0)\right] + \left[\cos(\pi) - \cos(0)\right]$

$= \left[0 - 0\right] + \left[-1 - (1)\right] = -2$

$u = x \quad dv = \cos(x)dx$

$du = dx \quad v = \sin(x)$

$\sin(\pi) = 0, \ \sin(0) = 0$

$\cos(\pi) = -1, \ \cos(0) = 1$

43. $\int_{-\pi}^{\pi} x \sin x \, dx$

$u = x$, $dv = \sin(x)\,dx$

$du = dx$, $v = -\cos(x)$

$= \left[-x\cos(x)\right]_{-\pi}^{\pi} - \int_{-\pi}^{\pi} -\cos(x)\,dx$

$= \left[-x\cos(x)\right]_{-\pi}^{\pi} + \left[\sin(x)\right]_{-\pi}^{\pi}$

$= \left[-\pi\cos(\pi) + (-\pi)\cos(-\pi)\right] + \left[\sin(\pi) - \sin(-\pi)\right]$

$\cos(\pi) = -1$, $\cos(-\pi) = -1$

$\sin(\pi) = 0$, $\sin(-\pi) = 0$

$= \left[-\pi(-1) + (-\pi)(-1)\right] + \left[0 - 0\right] = 2\pi$

45. $\int_{0}^{\pi/2} x^2 \sin x \, dx$

$u = x^2$, $dv = \sin(x)\,dx$

$du = 2x\,dx$, $v = -\cos(x)\,dx$

$= \left[-x^2\cos(x)\right]_{0}^{\frac{\pi}{2}} - \int_{0}^{\frac{\pi}{2}} -2x\cos(x)\,dx$

$u = 2x$, $dv = \cos(x)\,dx$

$du = 2dx$, $v = \sin(x)$

$= \left[-x^2\cos(x)\right]_{0}^{\frac{\pi}{2}} + \int_{0}^{\frac{\pi}{2}} 2x\cos(x)\,dx$

$= \left[-x^2\cos(x)\right]_{0}^{\frac{\pi}{2}} + \left[2x\sin(x)\right]_{0}^{\frac{\pi}{2}} - \int_{0}^{\frac{\pi}{2}} 2\sin(x)\,dx$

$= \left[-x^2\cos(x)\right]_{0}^{\frac{\pi}{2}} + \left[2x\sin(x)\right]_{0}^{\frac{\pi}{2}} + \left[2\cos(x)\right]_{0}^{\frac{\pi}{2}}$

$= \left[-\frac{\pi}{4}\cos\left(\frac{\pi}{2}\right) + 0\cos(0)\right] + \left[\pi\sin\left(\frac{\pi}{2}\right) - 2\cdot 0\cdot\sin(0)\right] + \left[2\cos\left(\frac{\pi}{2}\right) - 2\cos(0)\right]$

$\cos\left(\frac{\pi}{2}\right) = 0$

$\sin\left(\frac{\pi}{2}\right) = 1$

$= \left[0 + 0\right] + \left[\pi - 0\right] + \left[0 - 2\right] = \pi - 2$